



The 9.0 TMUA Handbook

first edition

For the Test of Mathematics for University Admission and the shared ESAT Mathematics 2 content.

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The 9.0 TMUA Handbook

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1 Orientation

The TMUA has a reputation for unpredictability, and after enough time with the papers we have come to think that reputation is undeserved. Sit down with every released paper, as we have done more times than we care to admit, and the opposite impression takes hold. The same ideas return year after year in slightly different clothes, the same wrong answers are laid out with the same care, and the same few minutes are lost in the same places. The test is not unpredictable at all. It is consistent to a degree that ought to be reassuring, because anything consistent can be prepared for.

Part of that consistency is forced on the examiners. The syllabus is capped below the frontier of A-level mathematics: no calculus of trigonometric or exponential functions, no product or chain rule, nothing a well-taught sixth former has not already met. Denied harder content, the setters make questions difficult in the only way left to them, which is disguise. A quadratic arrives dressed as an exponential equation. A straight line hides inside an iterated logarithm. A circle theorem you last saw in Year 11 turns up wearing coordinates. Strip the disguise and most questions run to two or three lines; fail to strip it and you can spend five honest minutes getting nowhere. The marks that separate a 6.0 from a 7.5 belong, almost without exception, to the candidates who recognise structure quickly and who decline a short list of invitations to go wrong that the papers have been extending for a decade.

We run tmua.fyi, and over the past few years we have watched thousands of candidates work through the released papers on the platform, question by question and minute by minute. You learn a great deal from that vantage point, and surprisingly little of it is about mathematics. You learn where strong students lose marks they had no business losing, which wrong options attract answer after answer, and what actually happens to accuracy when somebody refuses to abandon a question that is refusing them back. Those observations shaped this book: the order of the topics, the choice of every worked example, and the timing advice in Part III all follow from how marks are really won and lost, rather than from the specification's own ordering.

A word on how to read it. Part I treats Paper 1 as a set of recurring moves. Part II treats Paper 2, which most candidates underprepare and where the strongest quietly pull away. Part III is about time, which is the resource the exam is actually rationing. Part IV condenses everything into a trap index and a final-week card. The worked examples are original, and each one is built so that a single idea does the work. Please do attempt them cold before reading the solutions; the point of the attempt is to find out whether the idea occurred to you, and that is information no amount of reading can give you.

2 Priorities

If you take one page of this book seriously, make it this one. When we ask students where they expect to lose marks, they name the glamorous topics: hard integrals, strange sequences. When we watch where marks are actually lost, a rather different list emerges, and it has been stable for as long as we have been looking.

Paper 2 logic and proof. Scores here run well below Paper 1, on material whose entire content fits in a few pages. The reason is simply neglect. Past-paper practice drifts towards the mathematics students already enjoy, and the reasoning material looks too small to need attention. It is small. It is also, together with graph reasoning, where the strongest candidates separate themselves from the merely good ones, and a fortnight of short daily drills will move a Paper 2 score further than a month of extra algebra.

Trigonometry. The weakest large topic on Paper 1, and not because of the identities. The damage comes from three habits: missing the second triangle in the ambiguous case, dividing

away whole families of solutions, and simplifying $\sqrt{1 - \sin^2 \theta}$ without the modulus. All three are matters of discipline rather than knowledge, which is good news, because discipline is quick to install.

GCSE geometry. Circle theorems, congruence, quadrilaterals, solids. It is examined directly, it sits far below the level of everything else you are doing, and it is precisely where able A-level students go wrong, because nobody has looked at it for three years. One honest evening with a GCSE revision guide buys more marks than any other evening available to you, and we are aware of how unglamorous that sentence is.

The multi-statement format. Questions offering three Roman-numeral claims are answered faster than average and wrong more often, which tells you how the format is being treated and how it deserves to be treated. Three claims are three questions. Budget like it.

Inequalities and the behaviour of functions. Rarely the subject of a question, frequently its final step, and a steady leak of marks across every other topic. Ranges, admissible roots and the order of interval endpoints under a negative scaling recur so often that we have given them their own sections.

PART I

Paper 1: the transformation playbook

3 Bounded substitution

Paper 1 is fond of hiding a quadratic behind an exponential or trigonometric shell: $9^{\sin x}$, $a^{\cos x}$, symmetric expressions in a and b . By this stage of your preparation the substitution itself is the easy half, and nearly everyone makes it. The half that scores is what you do with the domain, because the new variable is not free. It inherits a range from the function that built it, and the wrong options are written, quite deliberately, from the answers produced by forgetting that.

Key point. When you set $y = g(x)$, write down the range of y before doing anything else. Extrema of the new expression may sit at the endpoints of that range rather than at the vertex; roots outside the range are not solutions; and Vieta's formulas, applied to the new equation, will cheerfully count roots the original problem does not have.

Worked example 3.1

Find the maximum value of $9^{\sin x} - 6 \cdot 3^{\sin x} + 10$ for real x .

Solution. Set $y = 3^{\sin x}$; since $\sin x \in [-1, 1]$, $y \in [\frac{1}{3}, 3]$, and

$$9^{\sin x} - 6 \cdot 3^{\sin x} + 10 = y^2 - 6y + 10 = (y - 3)^2 + 1.$$

The vertex sits at the right-hand endpoint $y = 3$, so the expression is decreasing on the whole range and the maximum is at $y = \frac{1}{3}$:

$$\left(\frac{1}{3} - 3\right)^2 + 1 = \frac{64}{9} + 1 = \frac{73}{9}.$$

The planted error. The value at the vertex, here 1, will be waiting for you among the options, and it is a double trap: it optimises over the wrong domain, and it answers a minimum when the question asked for a maximum.

It is worth pausing over what the setter has done here, because the pattern repeats. The

quadratic is deliberately gentle; there is nothing in $(y-3)^2+1$ to slow anyone down. The whole difficulty of the question has been moved into the two facts the substitution encourages you to forget: the range of y , and which end of the parabola is being asked about. In our experience, when the algebra in a TMUA question feels a size too easy for its position on the paper, that is the moment to slow down for five seconds and ask where the difficulty actually went.

Worked example 3.2

The difference between the maximum and minimum values of $a^{\sin x}$, where $a > 0$, is 2. Find the sum of the possible values of a .

Solution. The extreme values are a and a^{-1} , so $|a - a^{-1}| = 2$. The two cases give

$$a^2 - 2a - 1 = 0 \Rightarrow a = 1 + \sqrt{2}, \quad a^2 + 2a - 1 = 0 \Rightarrow a = \sqrt{2} - 1,$$

keeping the positive root of each. The sum is $2\sqrt{2}$.

The planted error. Vieta's formulas applied to $a^2 - 2a - 1 = 0$ give 2: the sum of both roots of the quadratic, one of which is negative and is therefore not a value of a at all. Vieta counts the roots of the polynomial, not the roots of the problem.

The pair of cases is not optional, and there is a pleasing structural check hiding in it. For $a > 1$ the maximum is a ; for $a < 1$ it is a^{-1} ; and the valid values of a therefore arrive in reciprocal pairs. Sure enough, $(1 + \sqrt{2})(\sqrt{2} - 1) = 1$. If a variant of this question asked for the product of the possible values rather than the sum, you could write down 1 without touching a quadratic, and even when the sum is wanted, noticing the reciprocal structure is a ten-second audit of the whole solution.

Worked example 3.3

Real numbers a, b satisfy $(a^3 + \frac{1}{b^3})(\frac{1}{a^3} - b^3) = \frac{3}{2}$. Find the least possible value of ab .

Solution. Expanding, the constants cancel, leaving an equation in $t = (ab)^3$ alone:

$$\frac{1}{t} - t = \frac{3}{2} \Rightarrow 2t^2 + 3t - 2 = 0 \Rightarrow t = \frac{1}{2} \text{ or } t = -2,$$

so ab is $2^{-1/3}$ or $-2^{1/3}$, and the least value is $-2^{1/3}$.

A two-variable expression that only ever mentions a and b through products and reciprocals of the same shape is a one-variable question about ab in disguise. Expand, let the cross terms cancel, and give the surviving quantity a name. Trying to solve for a and b separately is not merely slow; it is impossible, because the question never determines them individually, and the setters know perfectly well that a certain number of candidates will try.

4 The auxiliary object

Some questions cannot be attacked where they stand, because the given information never mentions the thing you want. The move that unlocks them is to build a second object, a second

equation or an auxiliary polynomial, on which the given information suddenly becomes usable. It feels like a conjuring trick the first few times; it is actually one idea applied twice.

Key point. When a functional equation relates $f(x)$ to $f(1/x)$ or $f(a-x)$, substitute to produce a second equation and solve the pair simultaneously for $f(x)$. When a polynomial is pinned at several points that fit a simple pattern, subtract the pattern: the difference is a polynomial whose roots you know completely.

Worked example 4.1

A function f , defined for all real $x \neq 0$, satisfies

$$f(x) + 2f\left(\frac{1}{x}\right) = 6x$$

for every such x . Find the sum of all real solutions of $f(x) = 2$.

Solution. Replacing x by $1/x$ gives $f(1/x) + 2f(x) = 6/x$. Doubling this and subtracting the original equation eliminates $f(1/x)$:

$$3f(x) = \frac{12}{x} - 6x \implies f(x) = \frac{4}{x} - 2x.$$

Then $f(x) = 2$ gives $4 - 2x^2 = 2x$, that is $x^2 + x - 2 = 0$, with roots 1 and -2 . Both are nonzero, hence admissible, and the sum is -1 .

Notice why the substitution is legitimate: $1/x$ ranges over exactly the same domain as x , so the relation really does hold at both points. One equation tying $f(x)$ to $f(1/x)$ can never determine f ; two can, and the second was free. Notice also the small check that both roots are nonzero. Had one of them been 0, the sum-of-roots shortcut would have quietly counted a point outside the domain, which is the same planted error we met in Section 3 wearing different clothes.

Worked example 4.2

A cubic polynomial p satisfies $p(1) = 2$, $p(2) = 4$, $p(3) = 6$ and $p(4) = 20$. Find $p(0)$.

Solution. The first three conditions say $p(x) - 2x$ vanishes at 1, 2, 3. Since p is cubic, so is $q(x) = p(x) - 2x$, and

$$q(x) = c(x-1)(x-2)(x-3)$$

for some constant c . The fourth condition fixes it: $q(4) = p(4) - 8 = 12$ and $(3)(2)(1) = 6$, so $c = 2$. Then

$$p(0) = q(0) + 0 = 2(-1)(-2)(-3) = -12.$$

The planted error. Forgetting to restore the subtracted pattern at the end. Here it is harmless, because $2x$ vanishes at 0; had the pinned values been $p(n) = 2n + 1$, evaluating $q(0)$ and calling it $p(0)$ would miss by exactly the pattern. Put the pattern back explicitly before evaluating anywhere.

The values 2, 4, 6 were not an accident of question design; they were chosen so that a low-

degree pattern would be visible to anyone who looked for one. That is worth internalising as a general expectation: three interpolation conditions do not determine a cubic, so if a question of this shape is answerable at all, the pinned values must fit a pattern. If you cannot see one, you have not looked long enough; on this examination it is always there.

5 Inequalities

Inequalities are rarely the destination on this paper. They are the final step of questions that were officially about discriminants, logarithms or graphs, and the marks leak in the mechanics. Two mechanical rules stop the leak, and both exist to make case analysis unnecessary, because on a timed paper avoidable case analysis is a tax you pay in minutes.

Key point. To clear a denominator of unknown sign, multiply by its square, which is never negative: $\frac{P(x)}{x+k} > Q(x)$ becomes $(x+k)P(x) > (x+k)^2Q(x)$ with the direction preserved, and $x = -k$ excluded at the end. To remove moduli from $|A| > |B|$, square both sides: both are non-negative, so squaring is valid and the inequality becomes polynomial. Squaring an inequality whose sides can be negative is never valid.

Worked example 5.1

Solve $\frac{3x-1}{x-2} \geq 2$.

Solution. Multiply by $(x-2)^2 \geq 0$:

$$(3x-1)(x-2) \geq 2(x-2)^2 \implies (x-2)[(3x-1)-2(x-2)] \geq 0 \implies (x-2)(x+3) \geq 0,$$

so $x \leq -3$ or $x \geq 2$; excluding the undefined point, $x \leq -3$ or $x > 2$. A single test value confirms the reading: at $x = 0$ the left side is $\frac{1}{2}$, correctly outside.

Worked example 5.2

Solve $|x-3| > 2|x|$.

Solution. Both sides are non-negative, so squaring preserves the inequality:

$$x^2 - 6x + 9 > 4x^2 \implies 3x^2 + 6x - 9 < 0 \implies x^2 + 2x - 3 < 0 \implies -3 < x < 1.$$

Sign-based case analysis reaches the same answers, in about four times the space, with two sign errors available along the way. We have watched a great many candidates take that longer road under time pressure, and the ones who arrive do not always arrive with the right interval. The habit worth building is to reach for the square first and to save the case split for the rare question that truly needs one.

Key point. If $x \mapsto x/k$ carries one inequality to another, the solution set scales by k , and $k < 0$ swaps the endpoints. A stated sign condition on a parameter is there to be used; if your solution never used it, the unflipped option has probably been chosen.

Worked example 5.3

The solution set of $x^2 + bx + c < 0$ is $p < x < q$. For $k < 0$, solve $\left(\frac{x}{k}\right)^2 + b\left(\frac{x}{k}\right) + c < 0$.

Solution. The inequality says $x/k \in (p, q)$; multiplying by $k < 0$ reverses both inequalities:

$$kq < x < kp.$$

6 Integrals: parity, reflection, staircases, signed area

The specification contains only the power rule for integration, and the examiners are as aware of that as you are. So the integrals worth setting are exactly the ones the power rule does not reach directly: integrands with a symmetry, integrands built from the floor function, and areas that are not integrals at all. Each has a one-move removal, and learning to spot which one applies is most of the topic.

Key point. Over an interval symmetric about the origin, an odd integrand gives 0 and an even one gives twice the half-range integral; read parity before reaching for an antiderivative. A sum of odd powers over a symmetric interval, however unintegrable it looks, is 0 on sight. Over $[0, a]$, the reflection $x \mapsto a - x$ preserves the integral, and adding the reflected copy collapses $\frac{g(x)}{g(x)+g(a-x)}$ to $\frac{1}{2}$. Floor and ceiling integrands are staircases: draw the steps and add rectangles.

Worked example 6.1

Evaluate $\int_0^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{3-x}} dx$.

Solution. Call the integral I and substitute $x \mapsto 3 - x$, which maps $[0, 3]$ onto itself:

$$I = \int_0^3 \frac{\sqrt{3-x}}{\sqrt{3-x} + \sqrt{x}} dx.$$

Adding the two forms, the numerators rebuild the common denominator, so $2I = \int_0^3 1 dx = 3$ and $I = \frac{3}{2}$.

The surd never gets integrated, and in fact it cannot be by any method on the specification. That is not an oversight; it is the hint. When an integrand is visibly beyond the syllabus, the examiners are not testing whether you know a technique they excluded. They are inviting you to notice a symmetry, and the invitation has a deadline.

Worked example 6.2

Evaluate $\int_0^5 \lfloor x \rfloor dx$.

Solution. The integrand is 0, 1, 2, 3, 4 on successive unit intervals, each a rectangle of width 1:

$$0 + 1 + 2 + 3 + 4 = 10.$$

There is nothing to antidifferentiate here, and trying is the mistake the question exists to catch. The general value $\int_0^n \lfloor x \rfloor dx = \frac{1}{2}n(n-1)$ is worth carrying, but the picture is worth more, because the variants with $\lfloor \sqrt{x} \rfloor$ or a ceiling change the widths of the rectangles and only the picture keeps up with them.

Worked example 6.3

Find the total area enclosed between $y = x^2 - 4$, the x -axis, and the lines $x = -3$, $x = 3$.

Solution. The crossings $x = \pm 2$ lie inside the range, so split there. By symmetry,

$$2 \left(\int_0^2 (4 - x^2) dx + \int_2^3 (x^2 - 4) dx \right) = 2 \left(\frac{16}{3} + \frac{7}{3} \right) = \frac{46}{3}.$$

The planted error. $\left| \int_{-3}^3 (x^2 - 4) dx \right| = 6$ will be among the options. The word “area” is an instruction: split the range at every axis crossing and add the magnitudes.

Of every plant in this book, the signed-integral option is the most dependable. It appears whenever a curve crosses the axis inside the limits, it is always the number a quick candidate computes, and we have seen it chosen by students who could explain the distinction perfectly well in a calm room. The defence is a habit rather than a piece of knowledge: the moment you read the word “area”, mark the axis crossings, before any integral sign gets written.

7 Discriminant thinking, completed

Whether two curves meet, and how many times, is a question about the sign of a discriminant, and it is answered without finding a single point of intersection. Reduce the meeting to one equation and read Δ : two solutions when $\Delta > 0$, tangency at $\Delta = 0$, none when $\Delta < 0$. If the leading coefficient contains the parameter, examine its vanishing separately, because a quadratic that has degenerated into a line is not tangent to anything. All of that you likely know. The refinement that decides the harder questions is that the discriminant only controls whether roots are real; plenty of questions also need to know where they are.

Key point. After a substitution $y = x^2$, $y = a^x$, $y = \sin x$, only roots inside the image of the substitution count. Two distinct positive roots of $y^2 + by + c$ means three conditions: $\Delta > 0$, $c > 0$, $-b > 0$.

Worked example 7.1

For which p, q does $x^4 - 2px^2 + q = 0$ have four distinct real roots?

Solution. Each distinct positive root of $y^2 - 2py + q = 0$ gives two real x ; zero or negative roots give fewer. Two distinct positive roots requires

$$p^2 > q, \quad q > 0, \quad p > 0,$$

and nothing shorter suffices.

The planted error. $p^2 > q$ on its own: with $q < 0$ one root in y is negative and only two real x survive.

Each of the three conditions is pulling real weight. The discriminant excludes repeated roots, $q > 0$ excludes a negative root, and $p > 0$ excludes the case of both roots negative. The wrong options for a question like this are manufactured by deleting one condition at a time, so if you know what each is for, you can identify which option corresponds to which omission. That turns checking your answer into a thirty-second exercise instead of a re-derivation, which matters late in the paper.

8 The modulus discipline

Key point. $\sqrt{x^2} = |x|$, and therefore $\sqrt{1 - \sin^2 \theta} = |\cos \theta|$ and $\sqrt{1 - \cos^2 \theta} = |\sin \theta|$. Simplifying the root of a square produces a modulus and a case split, and an answer obtained without the split is wrong on half the domain. The options are priced accordingly.

Worked example 8.1

For how many of $\theta = 1^\circ, 2^\circ, \dots, 360^\circ$ does $\sqrt{1 - \cos^2 \theta} = -\sin \theta$ hold?

Solution. The left side is $|\sin \theta|$, and $|\sin \theta| = -\sin \theta$ exactly when $\sin \theta \leq 0$: every θ from 180° to 360° inclusive, so 181 values.

The planted error. Dropping the modulus turns the equation into $\sin \theta = -\sin \theta$, which has two solutions in range instead of 181.

Take a moment over the scale of that difference. This is not a technicality that costs the odd mark at the margin; one dropped modulus turned a count of 181 into a count of 2, and counting questions are exactly where the identity tends to be deployed. Our rule of thumb: whenever a solution count on a trigonometric range comes out small and tidy, audit the root signs before you feel pleased about it.

9 Logarithms and exponentials

Equations in a^x hide quadratics. Iterated logarithms hide straight lines. Comparisons of enormous powers hide a common exponent. Three disguises, three removals, and between them they cover nearly everything this topic will ask of you.

Key point. In any equation featuring a^{2x} and a^x , substitute $y = a^x > 0$, solve the quadratic in y , and reject non-positive roots before recovering x . Substitute t for the innermost logarithm of an iterated expression: everything becomes affine in t , and comparisons become comparisons of lines. To rank b^m against c^n , rewrite both with a common exponent.

Worked example 9.1

Solve $4^x - 3 \cdot 2^{x+1} + 8 = 0$.

Solution. With $y = 2^x$, note $2^{x+1} = 2y$, so

$$y^2 - 6y + 8 = 0 \implies y = 2 \text{ or } y = 4,$$

both positive and admissible: $x = 1$ or $x = 2$.

Here both roots survived, which makes the admissibility check look like a formality. It earns its keep on the variants where one root of the quadratic comes out negative or zero, because there the number of solutions is the question, and the rejected root is the entire difference between the right option and its neighbour. This is the bounded-substitution rule of Section 3 again: $y = a^x$ has range $(0, \infty)$, and the range travels with the substitution.

Worked example 9.2

For $x > 16$, which is larger: $\log_2(\log_2 \sqrt{x})$ or $\frac{1}{2} \log_2(\log_2 x)$?

Solution. With $t = \log_2(\log_2 x)$,

$$\log_2(\log_2 \sqrt{x}) = \log_2\left(\frac{1}{2} \log_2 x\right) = t - 1, \quad \frac{1}{2} \log_2(\log_2 x) = \frac{t}{2}.$$

The lines cross at $t = 2$, that is at $x = 16$; for $x > 16$ we have $t - 1 > \frac{t}{2}$, so the first expression is larger.

Nested logarithms are the most reliably frightening objects on the paper, and the fright is doing all of the work, because after the substitution this question was about two straight lines. Candidates who try to reason about the nested forms directly almost always misplace the crossing point. Name the inner logarithm, find where the lines cross, translate that value back into x , and the ranking on either side reads off the slopes.

Worked example 9.3

Which is larger: 2^{300} or 3^{200} ?

Solution. Write both as hundredth powers:

$$2^{300} = (2^3)^{100} = 8^{100}, \quad 3^{200} = (3^2)^{100} = 9^{100},$$

so 3^{200} is larger.

Taking logarithms reaches the same comparison, but it leaves you holding $300 \log 2$ against $200 \log 3$, and on a calculator-free paper you do not have the values. The common exponent needs nothing: it converts a comparison of towers into a comparison of 8 and 9. Whenever two large powers must be ranked, look for the shared exponent first.

10 Trigonometry

Key point. The ambiguous case is settled by the height. Given the side b , the angle θ at one end of it, and the opposite side a : there is no triangle if $a < b \sin \theta$; exactly one, right-angled, if $a = b \sin \theta$; two if $b \sin \theta < a < b$; and one if $a \geq b$. Sketch the arc every time. The second triangle is usually the question.

Worked example 10.1

In a triangle, one side has length 8, the angle at one end of it is 30° , and the side opposite that angle has length a . For which values of a are there two possible triangles?

Solution. The height of the configuration is $8 \sin 30^\circ = 4$. Two triangles exist exactly when the opposite side clears the height but not the given side:

$$4 < a < 8.$$

At $a = 4$ the triangle is unique and right-angled; from $a = 8$ upwards the arc meets the base line only once.

The sine rule, used mechanically, delivers $\sin B$ and then falls silent; it does not volunteer that $180^\circ - B$ is also available. A sketch does. Draw the base, mark the angle, and swing an arc of radius a from the far end: the two intersections with the base line are the two triangles, and you can see whether there are two before computing anything at all.

Key point. Never divide by a trigonometric factor: collect everything on one side and factorise. Dividing by $\sin x$ discards the whole $\sin x = 0$ family, and the number of solutions is usually the question. Treat interval endpoints exactly as the stem gives them.

Worked example 10.2

How many values of x in $[0, 2\pi]$ satisfy $\tan x = 3 \sin x$?

Solution. Writing $\tan x = \frac{\sin x}{\cos x}$ and collecting,

$$\frac{\sin x (1 - 3 \cos x)}{\cos x} = 0,$$

so $\sin x = 0$ or $\cos x = \frac{1}{3}$. The first gives $x = 0, \pi, 2\pi$; the second gives two values, one in each of the first and fourth quadrants. Five solutions.

The planted error. Dividing through by $\sin x$ keeps only the two solutions of $\cos x = \frac{1}{3}$. The family thrown away is bigger than the one kept.

The third discipline for this topic lives back in Section 8: any appearance of $\sqrt{1 - \sin^2 \theta}$ is a case split, not a simplification. Between the arc, the refusal to divide, and the modulus, you have covered nearly everything trigonometry does to candidates on this paper, and none of the three takes longer than a breath to apply.

11 Graphs of functions

If we had to name one skill that most cleanly separates excellent candidates from good ones, it would be the willingness to answer questions with a shape rather than a computation. Three shapes do most of the work on this paper: the cubic with its two stationary values, the folded curve $|f(x)|$, and the transformed graph $f(ax + b)$.

Key point. A cubic has three distinct real roots exactly when its two stationary values have opposite signs. To count solutions of $f(x) = k$, draw $y = f(x)$ and slide the horizontal line $y = k$; the count changes only as k passes a stationary value or a fold. For $f(ax + b)$, factorise the argument as $a(x + \frac{b}{a})$: translate by $\frac{b}{a}$, then stretch by $\frac{1}{a}$, in that order.

Worked example 11.1

How many real roots does $x^3 - 3x + 1 = 0$ have?

Solution. The derivative $3x^2 - 3$ vanishes at $x = \pm 1$, with

$$f(-1) = 3, \quad f(1) = -1.$$

Opposite signs, so the curve crosses the axis three times: three real roots. No root is ever located, and none of them is pleasant to locate.

Worked example 11.2

For which values of k does $|x^2 - 4| = k$ have exactly three real solutions?

Solution. The graph of $|x^2 - 4|$ is the parabola with its dip folded upwards: zeros at $x = \pm 2$, a local maximum of 4 at $x = 0$. A horizontal line meets it four times for $0 < k < 4$, twice for $k = 0$ or $k > 4$, and exactly three times when it passes through the peak of the fold: $k = 4$, where the solutions are $x = 0$ and $x = \pm 2\sqrt{2}$.

“Exactly three” is a favourite of the setters, and for good reason: it happens only at the exceptional value, the line through the peak, so it cannot be found by trying typical values of k . There is also a symmetry argument worth having in your pocket: this graph is even, so solutions come in pairs except on the axis of symmetry, and an odd count therefore forces one solution at $x = 0$. Checking that thought against the options is sometimes faster than the sketch itself.

Worked example 11.3

The graph of $y = \cos x$ is transformed into $y = \cos(2x - \frac{\pi}{3})$ by a stretch parallel to the x -axis followed by a translation. Which translation?

Solution. Factorise the argument:

$$\cos(2x - \frac{\pi}{3}) = \cos\left(2\left(x - \frac{\pi}{6}\right)\right).$$

After the stretch by factor $\frac{1}{2}$ produces $\cos 2x$, the required translation is $\frac{\pi}{6}$ in the positive x -direction.

The planted error. Reading the translation as $\frac{\pi}{3}$ straight off the unfactorised argument. The factorisation is the entire content of the question, which is precisely why the argument is presented unfactorised.

12 Sequences and series

A sequence question wearing an alarming index, u_{2026} say, or the sum of eighty surd terms, is a periodicity or telescoping question in costume. Nobody is expected to iterate two thousand times, and the setters are relying on a certain number of candidates trying anyway.

Key point. For a recurrence with an unreachable index, compute terms until the cycle appears, then reduce the index modulo the period. For sums of differences in disguise (reciprocals of surd sums, partial fractions), rewrite the general term as $g(n+1) - g(n)$ and telescope. For geometric information given as two sums, take a ratio in which the first term cancels.

Worked example 12.1

A sequence has $u_1 = 3$ and $u_{n+1} = \frac{1}{1 - u_n}$ for $n \geq 1$. Find u_{2026} .

Solution. Iterating,

$$u_1 = 3, \quad u_2 = -\frac{1}{2}, \quad u_3 = \frac{2}{3}, \quad u_4 = 3,$$

so the sequence has period 3. Since $2026 = 3 \times 675 + 1$, $u_{2026} = u_1 = 3$.

Worked example 12.2

Evaluate $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{80} + \sqrt{81}}$.

Solution. Rationalising the general term,

$$\frac{1}{\sqrt{n} + \sqrt{n+1}} = \sqrt{n+1} - \sqrt{n},$$

so the sum telescopes to $\sqrt{81} - \sqrt{1} = 8$.

Worked example 12.3

A geometric series with first term a and ratio r , where $|r| < 1$, has sum to infinity 8. The series of squared terms has sum to infinity 32. Find a .

Solution. The squared terms are geometric with first term a^2 and ratio r^2 . Dividing the second sum by the square of the first,

$$\frac{a^2/(1-r^2)}{a^2/(1-r)^2} = \frac{1-r}{1+r} = \frac{32}{64} = \frac{1}{2} \implies r = \frac{1}{3}, \quad a = 8(1-r) = \frac{16}{3}.$$

The planted error. Two plants recur in this territory. Dividing the sum of squares by the first sum, rather than by its square, leaves a alive inside the ratio and leads to a confidently wrong number, which will of course be among the options. And in alternating recurrences such as $b_{n+1} = b_n + (-1)^n$, everything hangs on the first step: from $b_1 = 5$ the terms run 5, 4, 5, 4, ..., the first step going down, not 5, 6, 5, 6, ... Write out four terms before you sum anything.

13 Counting and probability

The probability content here is genuinely GCSE-level, and the questions stay examinable by forcing a single decision: ordered or unordered, direct or complement. Get the frame right and the arithmetic is trivial; get it wrong and your answer is out by a factor that has been helpfully anticipated among the options.

Key point. Count ordered selections with products of choices; count unordered selections with binomial coefficients; and never mix the two inside one fraction. Go to the complement when the event is “at least one”; count directly when it is “all different” or “exactly one”. A good bijection, pairing what you must count with something you can count, beats any formula.

Worked example 13.1

How many three-digit numbers have strictly increasing digits?

Solution. A strictly increasing string cannot contain 0, since 0 would have to lead. So the digits are three distinct choices from $\{1, \dots, 9\}$, and each choice has exactly one increasing arrangement:

$$\binom{9}{3} = 84.$$

The bijection is the entire solution: unordered choices of digits correspond one-to-one with valid numbers, so the count is a binomial coefficient and no arrangement factor ever appears. The options will offer you $\binom{10}{3}$, which forgot about the leading zero, and $9 \times 8 \times 7$, which counted ordered strings. Both are answers to nearby questions that were not asked.

Worked example 13.2

Three fair dice are rolled. What is the probability that the three scores are all different?

Solution. Count ordered outcomes on both levels:

$$\frac{6 \times 5 \times 4}{6^3} = \frac{120}{216} = \frac{5}{9}.$$

Direct counting wins here because “all different” is one clean pattern. Its complement, “at least two equal”, splinters into cases and invites the classic double-count of the triple-equal outcomes. The complement is a tool for events whose direct description splinters, not a reflex; choose it when it simplifies, and only then.

14 Geometry

We will say it once more, because it is the highest-value sentence in the book: revise the GCSE block. Circle theorems, congruence criteria, quadrilateral properties, solids. One evening. Nothing else pays like it. With that said, here are the shortcuts worth carrying and two configurations that recur.

Key point. Heron’s formula, $\sqrt{s(s-a)(s-b)(s-c)}$, gives area without angles. A chord at perpendicular distance d from the centre has length $2\sqrt{r^2 - d^2}$. The locus of points at which a segment AB subtends a right angle is the circle on AB as diameter, and tangency to a circle is the statement “distance from centre equals radius”. The shortest path along

a surface is a straight line in the unfolded net.

Worked example 14.1

A triangle has sides 7, 8, 9. Find its area.

Solution. $s = 12$, and Heron gives

$$\sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} = 12\sqrt{5}.$$

No angle was found. The cosine rule would deliver the same area after considerably more work, all of it error-prone under time pressure.

Worked example 14.2

$A(0, 0)$ and $B(8, 0)$ are fixed, and P moves so that $\angle APB = 90^\circ$. The line $y = x + c$ is tangent to the locus of P . Find the sum of the possible values of c .

Solution. By the converse of the angle-in-a-semicircle theorem the locus is the circle on AB as diameter: centre $(4, 0)$, radius 4. Writing the line as $x - y + c = 0$, tangency requires

$$\frac{|4 + c|}{\sqrt{2}} = 4 \implies c = -4 \pm 4\sqrt{2},$$

and the sum is -8 .

Two recognitions, chained: Thales turns the angle condition into a circle, and the distance formula turns tangency into a single equation. It is possible to do this question by parametrising P and imposing the right angle algebraically, in the same sense that it is possible to walk to the next town: you will arrive, eventually, at the same circle, having spent ten minutes you did not have.

Worked example 14.3

A cube has edge 6. Find the shortest path along its surface between opposite corners.

Solution. Unfold two faces into a 6×12 rectangle:

$$\sqrt{6^2 + 12^2} = 6\sqrt{5}.$$

The planted error. The space diagonal $6\sqrt{3}$ will be among the options. It is the distance through the solid, and it is what 3D coordinates on autopilot will hand you. “Along the surface” means the net.

PART II

Paper 2: where the top decile is decided

15 The economics of Paper 2

Here is the strange thing about Paper 2. The mathematics on it is, if anything, lighter than Paper 1's, and yet scores run lower. The weight sits in the reading: the difference between “if” and “only if”, between a statement and its converse, between the polynomial the question named and the cubic you happened to picture. These distinctions feel beneath the attention of a strong mathematician, and that feeling is exactly what makes them work as traps. Nobody loses these marks through inability. They lose them through speed.

We can be concrete about the behaviour, because we have watched it at scale: candidates spend less time on multi-statement reasoning questions than on ordinary ones, and get them wrong more often. The format reads quickly, each clause looks judgeable at a glance, and the glance is wrong about one clause in three. Whatever the format invites, a question with three claims is three questions, and it deserves three questions' worth of care.

The encouraging part: this material rewards drilling like nothing else on the test. The confusions are finite, the corrections are mechanical, and ten minutes a day for a fortnight on negation, converse and the direction of necessity will move a Paper 2 score further than any amount of additional algebra practice. We have watched that happen too.

16 Necessary and sufficient

You know the definitions; everyone does. The marks are lost in the traffic through them, at speed, in prose designed to blur the direction of the arrow. Two operational rules do the work.

Key point. Convert every phrasing to an arrow before you think: sufficient is $A \Rightarrow B$, necessary is $B \Rightarrow A$, and “ A only if B ” is $A \Rightarrow B$. Then use structure: in an “exactly one statement is true” question, two statements that are contrapositives of each other are the same statement, so neither can be the unique truth, and anything that implies a dead statement dies with it. Questions of this shape often collapse without a single fact about the function in the stem.

Worked example 16.1

y is a real number and g is a function. Exactly one of the following is true. Which?

- A) $y > 0$ only if $g(y) \leq 0$
- B) $y \leq 0$ if $g(y) > 0$
- C) $y > 0$ only if $g(y) > 0$
- D) $g(y) \leq 0$ if $y \leq 0$
- E) $g(y) > 0$ only if $y > 0$
- F) $g(y) > 0$ if and only if $y \leq 0$

Solution. As arrows: A is $y > 0 \Rightarrow g(y) \leq 0$ and B is its contrapositive; a dead pair. D is $y \leq 0 \Rightarrow g(y) \leq 0$ and E is its contrapositive; another dead pair. F implies B, and B is dead, so F is dead. Only C remains, and it must be the true one. Nothing about g was ever needed.

Look at the shape of that solution once more. No case analysis, no attempted examples, no properties of g ; the question was a logic crossword and the arrows were its grid. When we watch candidates struggle with this type, it is almost always because they translate one option at a time and reason about each in prose, which takes five minutes and ends in doubt. Translating all six as a block takes under a minute, and after that the elimination is mechanical. Translate first. Think second.

Worked example 16.2

f is a polynomial. P : $f(x) = 0$ for exactly three real x . Q : $f'(x) = 0$ for exactly two real x . Is P necessary and/or sufficient for Q ?

Solution. Neither. A quintic can cross the axis three times while f' vanishes four times, so $P \not\Rightarrow Q$; and a cubic with two stationary points whose local minimum sits above the axis has only one real root, so $Q \not\Rightarrow P$.

The planted error. For a cubic, $P \Rightarrow Q$ does happen to hold, and a cubic is the picture nearly everyone draws. When a stem names a wide class, polynomial, function, integer, the question lives at the edges of that class, and the planted option is the one that is true for the familiar special case.

Those two counterexamples are worth keeping as stock parts: the quintic with extra wiggles, and the cubic whose local minimum floats above the axis. Between them they demolish most careless claims connecting root counts to stationary-point counts, in either direction, and some costume of that claim appears most years.

17 The multi-statement discipline

Key point. Settle each numeral on its own. Try first to kill it with a small, ugly witness: a constant function, $f(x) = -x$, an even function, a rectangle, $n = 1$. Only if it survives do you attempt the one-line proof. Map to the option letters last. One settled numeral typically halves the options, so partial knowledge converts cleanly into marks; one guessed numeral poisons all three.

Worked example 17.1

f and g are cubics with leading coefficients a and p , quadratic coefficients b and q , linear coefficients c and r , constants d and s . Given $f(x) - g(x) > 0$ for all real x , which are necessarily true?

- I) $a > p$
- II) if $b = q$ then $c = r$
- III) $d > s$

Solution. $h = f - g$ is everywhere positive. If $a \neq p$ then h is a genuine cubic, and every cubic is unbounded below; impossible. So $a = p$, and statement I fails: the hypothesis forces equality, not inequality. With $a = p$ and $b = q$, h reduces to a linear function that is everywhere positive, hence constant: $c = r$, and II holds. $h(0) > 0$ gives III at once. II and III only.

The planted error. Reading “bigger everywhere” as “bigger leading coefficient”. An everywhere-positive difference acts on the leading terms first, because odd-degree terms are unbounded in both directions, and the numeral about leading coefficients is exactly where such questions bury the trap.

Notice, too, the order in which the solution took the numerals: not the printed order. III was a one-line evaluation, I was a degree argument, and II leaned on what I had already established. The printed order of numerals is chosen by the same people who chose the distractors; reorder by cost, cheapest first, and let each settled numeral prune the options before you pay for the expensive one.

Worked example 17.2

$f(x) = x^3 - 3qx^2 + 1$ has three distinct real roots. Which of these necessarily have three distinct real roots?

- I) $f(x) - 1 = 0$
- II) $f(x - 2) = 0$
- III) $-f(x) = 0$

Solution. II and III preserve the zero set trivially. For I, use stationary values: $f' = 3x(x - 2q)$ gives

$$f(0) = 1, \quad f(2q) = 1 - 4q^3,$$

and the hypothesis forces $1 - 4q^3 < 0$. Shifting down by 1 makes the stationary values 0 and $-4q^3$: one of them is exactly zero, so the shifted curve is tangent to the axis and

carries a repeated root. I necessarily fails; II and III only.

The planted error. Judging vertical shifts by intuition. Horizontal shifts and reflections never change a root count; a vertical shift always puts it in question; and the product of the shifted stationary values is the entire answer, in two lines, every time.

Worked example 17.3

Let $a > 1$. Must $a^{2x} = x^2$ have the same number of real solutions as $a^x = x$?

Solution. No. Since $a^{2x} = (a^x)^2$ and $a^x > 0$, the equation says $a^x = |x|$: it keeps every solution of $a^x = x$ and adds the solutions of $a^x = -x$, of which there is always exactly one, an increasing function crossing a decreasing one. So the counts always differ by one. Contrast $a^{2x} = 2x$, which the substitution $t = 2x$ maps solution-for-solution onto $a^t = t$: bijective substitutions preserve counts; squarings add branches.

Worked example 17.4

f is a polynomial and, for $p < q$,

$$J_{p,q} = \int_p^q (f(|x|) - f(x)) \, dx.$$

Which of the following must be true?

- I) If $J_{p,q} = 0$ then $p \geq 0$.
- II) If f is decreasing on all of $\mathbb{R}$ then $J_{p,q} \leq 0$ for all $p < q$.
- III) If $J_{p,q} > 0$ then $p < 0$.

Solution. For $x \geq 0$ the integrand is identically zero, and that single observation settles III at once: if $p \geq 0$ the whole range is dead and $J = 0$, so a positive J forces $p < 0$. The same thought kills I, with a witness: for an even polynomial, $f(|x|) = f(x)$ everywhere, so $J = 0$ for every $p < q$, including $p < 0$. For II: on $x < 0$ we have $|x| = -x > x$, and a decreasing f sends the larger argument to the smaller value, so the integrand is at most zero there and vanishes elsewhere: $J \leq 0$. II and III only.

This example repays the witness-first habit twice: the even polynomial disposes of I in a line, and “the integrand dies on $x \geq 0$ ” settles III before any real work begins. Constructions built on $f(|x|)$ or $|f(x)|$ always split the line into a dead half and a live half. Find the dead half first, and half the numerals tend to fall out of it.

18 Negation and quantifiers

This is the smallest section of the specification, and the money it costs candidates is out of all proportion to its size. Here is the entire calculus, followed by the two places it actually goes wrong.

Key point. A negated universal is existential: the negation of “every prime greater than 2 is odd” begins “some prime...”, never “every...”. The negation of $P \Rightarrow Q$ is $P \wedge \neg Q$: the antecedent survives un-negated, even when it is itself a conjunction. And $\forall x \exists y$ is weaker than $\exists y \forall x$: the first allows y to depend on x . Over the positive integers, remember the least element; it defeats careless universal claims.

Worked example 18.1

Write down the negation of: “for every positive integer n , if n is prime and $n > 2$, then n is odd.”

Solution. Flip the quantifier and negate the implication, leaving the antecedent intact:

there exists a positive integer n such that n is prime, $n > 2$, and n is even.

The planted error. Two corruptions of the antecedent appear among the options every time this is asked: De Morgan applied to it (“ n is not prime or $n \leq 2$, and...”) and the whole thing re-wrapped as another conditional (“if n is prime and $n > 2$ then n is not odd”). A quick self-test that catches both: the negation of an implication is a conjunction, so if what you have written still contains an “if”, it is wrong.

Worked example 18.2

Every variable ranges over the positive integers. Which of these are true?

- I) $\forall n \exists m (m > n)$
- II) $\exists m \forall n (m > n)$
- III) $\exists m \forall n (n \geq m)$

Solution. I is true: for each n , take $m = n + 1$; the witness is allowed to depend on n . II is false: it demands a single m larger than every positive integer. III is true: $m = 1$ works, because 1 is the least element. I and III.

Statements I and II differ only in the order of their quantifiers, and one is true while the other is false. That is the entire lesson, and the examiners retell it every year or two with a new interior. The quiet worker in III is the least element: over the integers, or the reals, III would fail, so the line that tells you the domain is load-bearing and deserves the same care as the quantifiers themselves.

19 Proof, and the errors planted in it

Error-spotting questions present a short argument that reaches a false or suspicious conclusion and ask for the first faulty step. The faults come from a small catalogue, and the step that carries one is always surrounded by steps that are deliberately sound, so that the argument fails in exactly one place.

Key point. Using the converse: the argument needs $P \Rightarrow Q$ and establishes $Q \Rightarrow P$, at exactly one step. Cancelling a possibly-zero factor: a chain of equalities that cancels a bracketed difference is hiding a division by zero. Squaring an equation admits strangers; squaring an inequality is valid only when both sides are non-negative. $\sqrt{x^2} = |x|$. And a non-monotonic function applied to both sides preserves nothing.

Worked example 19.1

A student argues as follows. “Let $x = y$. Step 1: $x^2 = xy$. Step 2: $x^2 - y^2 = xy - y^2$. Step 3: $(x + y)(x - y) = y(x - y)$. Step 4: $x + y = y$. Step 5: since $x = y$, $2y = y$, so $2 = 1$.” Which step is invalid?

Solution. Step 4 divides both sides by $x - y$, which the opening line makes zero. Steps 1 to 3 are honest algebra, and step 5 is honest given step 4. The flaw is the cancellation of a bracketed difference that the hypothesis forces to vanish.

Worked example 19.2

A student is asked to prove: if $27q^2 + 4p^3 < 0$, then $x^3 + px + q = 0$ has three distinct real roots. The argument runs: “Step 1: the stationary points are at $x = \pm\sqrt{-p/3}$, which exist since the hypothesis forces $p < 0$. Step 2: the product of the two stationary values is $q^2 + \frac{4p^3}{27}$, which is negative by hypothesis, so one stationary value is positive and one negative. Step 3: if a cubic has three distinct real roots, then one stationary value is positive and one is negative. Step 4: hence the equation has three distinct real roots.” Which step is invalid?

Solution. Step 3 is the converse of what step 4 needs. The chain requires “opposite-sign stationary values imply three distinct roots”; step 3 establishes the reverse implication, and step 4 then uses it as though it pointed forward. Steps 1 and 2 are watertight, which is typical of the genre: the computation is made sound so that the single reversed arrow is the only candidate.

The planted error. There is nearly always an option about the order or presentation of the steps, and it is nearly always the most popular wrong answer. It is a decoy. These questions are never about arrangement; read each step as an arrow, check its direction, and only if every arrow survives should you start considering missing cases.

If you want a drill that builds this skill quickly, try the following with a friend: each of you takes a correct proof from your notes, corrupts one step into its converse, and challenges the other to find it. It sounds like a parlour game. It is also precisely the skill under examination, and a week of it will make the real questions feel slow.

20 Construction and small-case questions

“Can the student draw a triangle such that...”, “for which n does this property force...”: questions like these reward one habit above all others, which is going hunting for the degenerate or

lopsided witness before trusting any pattern you observed in symmetric cases.

Key point. Test claims about triangles on a very obtuse one, never an equilateral one. Test claims about polygons on a non-square rectangle. Test claims about all n on $n = 1$ and on the first n where new behaviour can appear. Two generators produce most geometric counterexamples: $\sin \theta = \sin(180^\circ - \theta)$, which lets unequal angles have equal sines, and the room around an obtuse vertex, which admits perpendiculars that acute configurations forbid.

Worked example 20.1

A polygon with $n \geq 3$ vertices is inscribed in a circle whose centre lies inside it, and the radii to its vertices cut it into n triangles of equal area. Is the polygon necessarily regular?

Solution. Only for $n = 3$. Each triangle has area $\frac{1}{2}r^2 \sin(\text{central angle})$, so equal areas force equal sines of the central angles, and nothing more. A non-square rectangle, with central angles $\alpha, 180^\circ - \alpha, \alpha, 180^\circ - \alpha$, has equal areas without being regular, which settles $n = 4$ and generalises upwards. For $n = 3$, the sine conditions together with the angle sum of 360° force all three central angles to 120° : equilateral.

The planted error. “Equal areas, therefore congruent triangles, therefore regular.” The first link is false, and it is false by the supplementary-sine identity, which is why that identity leads the key point above.

This pattern extends well beyond circles. Whenever a symmetric conclusion (regular, equilateral, equal) is being offered from a symmetric-sounding hypothesis (equal areas, equal perimeters), our advice is to assume the setters have already built the lopsided counterexample, and to go looking for it before you attempt any proof. The right first instinct on “is it necessarily regular?” is not to prove but to wreck. And when the wrecking fails in an instructive way, so that you can see why it must fail, the proof usually assembles itself out of the wreckage, exactly as in the $n = 3$ case, where trying to keep the sines equal without the angles equal runs straight into the angle-sum constraint.

PART III

The time doctrine

21 The three-minute rule

Let us tell you what we have observed, because it is not what most candidates believe. Accuracy on a question is essentially flat from half a minute up to about two minutes of work. It drifts down through the third minute. And beyond four minutes, an answer is scarcely more reliable than a considered elimination-and-guess, however productive the grinding feels from the inside. The marginal minute past minute three buys approximately nothing, and its price is a whole other question somewhere later in the paper. The strongest candidates we watch are not conspicuously faster than the rest. They are simply the ones who refuse the fourth minute.

It helps to understand why the fourth minute is worthless, because understanding makes the rule easier to obey when it matters. A question that has resisted three minutes of a well-prepared candidate is nearly always resisting for one of two reasons. Either the recognition it wants has not fired, and grinding is not the operation that produces recognitions; or an early error is quietly poisoning everything downstream, and more of the same working compounds it. Both conditions respond to leaving and returning. Fresh eyes re-read the stem, and the stem is where the recognition lives.

Key point. At three minutes, stop and classify. Either the path to the answer is visible and short, in which case finish it; or it is not, in which case eliminate what you can, commit to an answer, flag the question, and move on. There is no negative marking: the flagged guess costs nothing, and flagged questions fall to fresh eyes far more often than they fall to the fourth minute.

22 The endgame

The hardest questions on both papers cluster in the final third, and the final third is exactly where time runs out. The result is a compounding problem we see constantly: the questions that need the most attention receive the least of it, taken at pace, by a tired candidate, and the traps of Part II convert best against exactly that person. The paper's ordering is, in this sense, adversarial, and you should refuse to let it dictate your spending.

Key point. Bank fifteen, then triage five. Take questions 1 to 15 at a target pace of about two minutes each, under the three-minute rule, strictly; unforced errors in that stretch are the most expensive on the paper. Spend the remaining time on 16 to 20, triaged by option structure rather than by order: multi-statement questions first, if you have trained Part II, because each settled numeral halves the options; pure computations last.

The triage order deserves a sentence of defence, because it runs against instinct. Late-paper multi-statement questions look the most frightening and are reliably the best value, since partial progress converts into eliminated options at a rate no computation matches. A late grind, by contrast, pays nothing until it pays everything, and with six minutes left, “nothing until everything” is a bet you cannot cover. Fear is not a pricing mechanism. Option structure is.

23 Options, guessing, review

The printed options are data, and reading them before you commit to a method is not cheating; it is the format, used as designed. Three readings recur. Growth: the behaviour of a candidate expression as $x \rightarrow 0$ or $x \rightarrow \infty$ disqualifies options wholesale, before any algebra. Parity and last digits: when the answer is an integer, its parity or final digit is often computable far faster than the integer itself, and one such feature usually leaves a single option standing. Back-substitution: when the options are concrete numbers, testing them in the original equation turns a solve into a check, and since exactly one option is correct, the first that works is the answer.

While we are here, let us retire a piece of folklore. Correct answers carry no positional bias worth knowing about, and “none of them” options are keyed correct at about the rate their share of the list would predict. There is no secret to extract from the letters. Choose the none option when the mathematics says so, and not otherwise; superstition about options is a tax paid by candidates who have run out of mathematics, and the remedy is usually one more elimination.

Key point. Review beats volume, and it is not close. Re-attempting the questions you previously missed, cold, before re-reading their solutions, improves scores faster per hour than sitting fresh papers. A paper sat and never reviewed is half a paper. The running list of your own missed questions is the syllabus for your final fortnight.

24 The final fortnight

The last two weeks are for consolidation, and the protocol is short enough to fit on a card.

Keep a miss list. Every question you get wrong in practice goes on one list, with a one-line note of the trap or the recognition involved: the one line, not the full solution. Re-attempt the list cold every few days, and retire an item only after two consecutive clean solves. The list converges faster than you expect, and what remains at the end is a personal trap index more valuable than any general one, this book’s included.

Sit mocks under match conditions. Two papers a week, timed, no pauses, every question answered, reviewed the same day. The review is the session; the sitting is just how you generate material for it. Every miss goes on the list, and every flagged-and-guessed question gets thirty seconds of honesty about whether the flag was the right call. You are practising the three-

minute rule at least as much as the mathematics; scores are a by-product, and for the first week you should expect them to wobble while the discipline beds in.

Rebuild arithmetic fluency. The paper is calculator-free, and the long division you have not done since Year 9 will happen, with six minutes left, whether you have practised it or not. Ten minutes a day on mental fractions, surd simplification and two-digit multiplication is unglamorous, and it repays itself on every single question of both papers.

The day before. The final-week card in Part IV, your own miss list, and nothing new. New material on the eve of an examination adds anxiety and no marks; recognition drills add both calm and marks. Then sleep, which is worth more than either.

PART IV

The trap index and the final-week card

25 The trap index

The planted error	The counter-move
The signed integral where area is asked	Split at every axis crossing; add magnitudes.
$\sqrt{1 - \sin^2 \theta}$ without the modulus	$\sqrt{x^2} = x $; case-split on the sign.
A substituted quadratic optimised over all reals	Carry the range of $y = g(x)$; check endpoints.
Vieta over roots outside the substitution's range	Admit roots one at a time; sum the survivors.
Discriminant alone for "four distinct real roots"	Two distinct positive roots: Δ , sum, product.
A scaled solution set with the endpoint order kept	Multiplying an interval by $k < 0$ swaps its endpoints.
Dividing an equation by a trigonometric factor	Collect and factorise; the discarded family is the question.
The second triangle of the ambiguous case missed	Compare the opposite side with $b \sin \theta$; sketch the arc.
A translation read off an unfactorised argument	$f(ax + b) = f(a(x + \frac{b}{a}))$: factorise first.
A recurrence iterated instead of cycled	Find the period; reduce the index modulo it.
An alternating sequence summed from the wrong first step	Write four terms before summing anything.
Ordered and unordered counts mixed in one fraction	Products for ordered, binomials for unordered; never both.
"Polynomial" read as "cubic"	Test the widest members of the named class.
Vertical shifts of a cubic judged by intuition	Both stationary values; the product's sign decides.
"Bigger everywhere" as "bigger leading coefficient"	Everywhere-positive differences kill odd leading terms.
Squaring without the modulus branch	$a^{2x} = x^2$ means $a^x = x $: two branches.
Equal areas read as equal angles	$\sin \theta = \sin(180^\circ - \theta)$; try the rectangle.
A negation that still contains "if"	$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$; antecedent intact.
The converse of one step, mid-proof	Read each step as an arrow; check its direction.
Step-order and style decoys in error-spotting	The flaw is always logical, never presentational.
The interior distance for a surface path	Unfold the net; measure in the plane.

26 The final-week card

1. Odd integrand, symmetric interval: zero. Even: double the half.
2. "Area": split at every crossing; the signed value is planted.
3. Substitute, then carry the range; optimise over the range.
4. After a substitution, admit roots one by one; Vieta counts phantoms.
5. Real roots are not placed roots: sum and product signs too.
6. $\sqrt{x^2} = |x|$; a dropped modulus halves the domain.
7. Scaling a solution set by a negative number reverses it.
8. Clear unknown-sign denominators with a square; square only non-negative sides.
9. Name the inner logarithm; iterated logs are lines in t .
10. Compare power towers through a common exponent.

11. Ambiguous case: compare the opposite side with $b \sin \theta$.
12. Never divide by a trigonometric factor; collect and factorise.
13. Factorise the argument before reading off a transformation.
14. A cubic's root count is the sign of the product of its stationary values.
15. Reduce impossible indices modulo the period; telescope sums of differences.
16. Ordered counts by products, unordered by binomials; never both at once.
17. Circle theorems and congruence: GCSE marks at A-level prices.
18. Shortest surface path: unfold the net.
19. Sufficient: $A \Rightarrow B$. Necessary: $B \Rightarrow A$. "Only if": $\Rightarrow$.
20. Contrapositive twins among the options eliminate each other.
21. "Polynomial" is not "cubic".
22. Multi-statement: three verdicts, counterexample first, letters last.
23. A negation never contains "if"; the antecedent survives untouched.
24. In a purported proof, find the arrow pointing backwards.
25. Three minutes, then classify: finish, or eliminate, answer, flag, move.

27 Appendix: the compressed formula sheet

Discriminant	$\Delta = b^2 - 4ac$; tangency at $\Delta = 0$
Two distinct positive roots	$\Delta > 0$, product > 0 , sum > 0
Vieta (quadratic)	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$
Vieta (cubic)	$\sum \alpha = -\frac{b}{a}$, $\sum \alpha\beta = \frac{c}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$
Sum of squares	$\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2 \sum \alpha\beta$
Logarithms	$\log_a x + \log_a y = \log_a xy$; $\log_a b = \frac{1}{\log_b a}$
Modulus	$\sqrt{x^2} = x $; $\sqrt{1 - \sin^2 \theta} = \cos \theta $
Identities	$\tan \theta = \frac{\sin \theta}{\cos \theta}$; $\sin^2 \theta + \cos^2 \theta = 1$
Supplementary sines	$\sin \theta = \sin(180^\circ - \theta)$
Ambiguous case	two triangles iff $b \sin \theta < a < b$
Triangle area	$\frac{1}{2}ab \sin C$; Heron: $\sqrt{s(s-a)(s-b)(s-c)}$
Chord	length $2\sqrt{r^2 - d^2}$ at distance d from the centre
Thales	the right-angle locus over AB : the circle on AB as diameter
Interior angle	regular n -gon: $\frac{180(n-2)}{n}$ degrees
Transformations	$f(ax+b) = f(a(x+\frac{b}{a}))$: translate $\frac{b}{a}$, then stretch $\frac{1}{a}$
Sums	$1 + 2 + \dots + n = \frac{1}{2}n(n+1)$; $\int_0^n [x] dx = \frac{1}{2}n(n-1)$
Geometric series	$S_\infty = \frac{a}{1-r}$ for $ r < 1$
Counting	ordered: products; unordered: $\binom{n}{r}$
Cubic root count	three distinct roots iff the stationary values have negative product
Negation	$\neg(\forall x P) \equiv \exists x \neg P$; $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
Conditional	equivalent to its contrapositive; not to its converse